RAJALAKSHMI ENGINEERING COLLEGE

M R Sally Madhulika 250701633-2025-CSE

M2

Started on	Friday, 11 September 2026, 8:59 AM
State	Finished
Completed on	Friday, 11 September 2026, 9:05 AM
Time taken	6 mins 2 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?

Falling back to raw text area.

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int count[1000] = {0};
    int x;

    for (int i = 0; i < n; i++) {
        scanf("%d", &x);

        count[x]++;

        if (count[x] == 2) {
            printf("%d", x);
            break;
        }
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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M R Sally Madhulika 250701633-2025-CSE

M2

Started on	Friday, 11 September 2026, 9:05 AM
State	Finished
Completed on	Friday, 11 September 2026, 9:16 AM
Time taken	10 mins 15 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, a[1000];
    scanf("%d", &n);

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[i] == a[j]) {
                printf("%d", a[i]);
                return 0;
            }
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

M R Sally Madhulika 250701633-2025-CSE

M2

Started on	Friday, 11 September 2026, 9:16 AM
State	Finished
Completed on	Friday, 11 September 2026, 9:26 AM
Time taken	9 mins 56 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2;
        scanf("%d", &n1);
        int a[n1];
        for (int i = 0; i < n1; i++){
            scanf("%d", &a[i]);
        }
        scanf("%d", &n2);
        int b[n2];
```

```
        for (int i = 0; i < n2; i++){
            scanf("%d", &b[i]);
        }
        int i = 0, j = 0;
        int first =1 ;
        while (i < n1 && j < n2) {
            if(a[i]==b[j]){
                if (!first)
                    printf(" ");
                printf("%d", a[i]);
                first = 0;
                i++;
                j++;
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }
        printf("\n");
    }
    return 0;
}

```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

M R Sally Madhulika 250701633-2025-CSE

M2

Started on	Friday, 11 September 2026, 9:29 AM
State	Finished
Completed on	Friday, 11 September 2026, 9:40 AM
Time taken	10 mins 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int t;
    if (scanf("%d", &t) == 1) {
        while (t--) {
            int n1, n2;
            scanf("%d", &n1);
            int *arr1 = (int *)malloc(n1 * sizeof(int));
            for (int i = 0; i < n1; i++) {
                scanf("%d", &arr1[i]);
            }
            scanf("%d", &n2);
```

```
            int *arr2 = (int *)malloc(n2 * sizeof(int));
            for (int i = 0; i < n2; i++) {
                scanf("%d", &arr2[i]);
            }

            int i = 0, j = 0;
            int first = 1;
            while (i < n1 && j < n2) {
                if (arr1[i] == arr2[j]) {
                    if (!first) printf(" ");
                    printf("%d", arr1[i]);
                    first = 0;
                    i++;
                    j++;
                } else if (arr1[i] < arr2[j]) {
                    i++;
                } else {
                    j++;
                }
            }
            printf("\n");
            free(arr1);
            free(arr2);
        }
    }
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

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Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

M R Sally Madhulika 250701633-2025-CSE

M2

Started on	Monday, 5 October 2026, 8:28 AM
State	Finished
Completed on	Monday, 5 October 2026, 8:30 AM
Time taken	2 mins 29 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n, k;
    scanf("%d", &n);

    int *A = (int *)malloc(n * sizeof(int));

    for (int i = 0; i < n; i++)
        scanf("%d", &A[i]);

    scanf("%d", &k);

    int i = 0, j = 1;

    while (i < n && j < n) {
        if (i == j) {
            j++;
        }
        else if (A[j] - A[i] == k) {
            printf("1");
            free(A);
            return 0;
        }
        else if (A[j] - A[i] < k) {
            j++;
        }
        else {
            i++;
        }
    }
```

```
33
34      printf("0");
35      free(A);
36
37      return 0;
38  }
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

M R Sally Madhulika 250701633-2025-CSE

M2

Started on	Monday, 5 October 2026, 8:34 AM
State	Finished
Completed on	Monday, 5 October 2026, 8:36 AM
Time taken	1 min 30 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, k;
    scanf("%d", &n);

    int A[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &A[i]);

    scanf("%d", &k);

    int i = 0, j = 1;

    while (i < n && j < n) {
        if (i == j)
            j++;
        else if (A[j] - A[i] == k) {
            printf("1");
            return 0;
        }
        else if (A[j] - A[i] < k)
            j++;
        else
            i++;
    }

    printf("0");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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